MEDTOUREASY — 3-TIER INTERVIEW PREPARATION STRATEGY

Resume-Aligned Scripts (30-40 Sec | 2-3 Min | 10-15 Min Deep-Dive) & Audited Metric Cheat Sheet

YOUR EXACT RESUME ENTRY (SOURCE OF TRUTH FOR INTERVIEWERS)

Data Science & Analytics Intern | MedTourEasy (May'26 – July'26)

- [Bullet 1] Engineered Python/SQL pipeline across 1,500 patient journeys and 5 datasets for funnel and provider analytics.
- [Bullet 2] Quantified a 16.7 pp SLA conversion gap (62.7% vs. 79.5%) using Chi-square statistical testing ($p < 0.001$).
- [Bullet 3] Built Logistic Regression and Random Forest classifiers, achieving 0.8695 ROC-AUC with leakage-aware features.
- [Bullet 4] Segmented 1,500 patients into 3 personas using K-Means on intent, engagement, response time, and case value.

MASTER NUMBERS CHEAT SHEET (MEMORIZE COLD BEFORE ENTERING THE INTERVIEW)

- Cohort & 5 Datasets: 1,500 inquiries | 554 completed (36.93%) | 350 explicit drop-offs (23.33%) | 596 stagnant open leads. Tables: patient_journey (1,500), provider_master (30), communication_logs (6,025), treatment_outcomes (554), country_reference (12). Zero PK/FK orphans.
- Audited Financials: INR 248,325,245 Realized Revenue (INR 448.2k/case) | INR 136,810,889 Cost | INR 111,514,356 Gross Margin (44.91% margin; INR 201.3k/case).
- Critical Funnel Anomaly: 229 of 554 completed patients (41.34%) had quote_shared=0, inflating naive quote-to-treatment conversion to 98.75% (554/561) vs. True Strict Sequential conversion of 57.93% (325/561).
- SLA Proof: 82.80% breached country SLA (1,242/1,500). Consult booking: 79.46% (SLA Met) vs 62.72% (Breached) -> -16.74 pp (Chi2=25.78, p=3.83e-7). Welch t-test: 15.77h (booked) vs 19.40h (unbooked), p=4.84e-15.
- ML & Personas: 72h cutoff -> Random Forest 0.8695 ROC-AUC vs Logistic Regression 0.8193. K-Means (k=3) isolates 426 High-Value At-Risk leads (INR 716.8k avg value, 31.9% conv). 10% drop-off recovery = INR 16.65M rev / INR 7.48M margin.

STRATEGY 1: THE 30–40 SECOND ELEVATOR PITCH (WORD-FOR-WORD SCRIPT)

When to use: Opening intro, 'Walk me through a project on your resume', or Recruiter/HM screen (~105 words).

"At MedTourEasy, I built an end-to-end patient conversion and provider intelligence system across 1,500 international medical tourism inquiries and 5 relational datasets representing INR 248.3 million in completed treatment revenue and INR 111.5 million in gross margin.

First, I engineered a Python and SQL data quality pipeline that uncovered 229 off-system treatment completions that were artificially inflating quote conversion from 57.9% to 98.8%. Second, using Chi-square testing against country-specific SLA benchmarks, I proved that 82.8% of inquiries breached SLA, causing a 16.7 percentage point drop in consultation booking (62.7% vs. 79.5%, $p < 0.001$).

Finally, I built a leakage-free Random Forest model achieving 0.8695 ROC-AUC at a 72-hour prediction cutoff and combined it with K-Means clustering (k=3) to isolate 426 High-Value At-Risk patients averaging INR 716.8k per case—creating a targeted routing playbook to recover an estimated INR 7.5 million in gross margin at a 10% recovery rate."

DELIVERY CUES & WHY THIS 30–40 SECOND SCRIPT WORKS

- [00–10 sec] Business Scale Hook: Anchor the scale immediately (1,500 inquiries, 5 datasets, INR 248.3M revenue, INR 111.5M margin).
- [10–25 sec] Engineering + Statistical Rigor: Hit the 229 unquoted completions anomaly (57.9% vs 98.8%) and the -16.7 pp SLA Chi-square result ($p < 0.001$).
- [25–40 sec] ML + Personas + ROI: Pair the 0.8695 ROC-AUC model (72h cutoff) with the 426 High-Value At-Risk K-Means persona and INR 7.5M margin recovery.
- Strategic Pause at 0:40: Stop right after 'INR 7.5 million in gross margin.' This invites the interviewer to ask about your 72h leakage guardrails or the 229 unquoted completions.

STRATEGY 2: THE 2–3 MINUTE STRUCTURED WALKTHROUGH

Hiring Manager & Technical Screen Script — Mapped 1:1 to Your 4 Resume Bullets (~400 Words)

PART 1 (0:00 – 0:45) | BUSINESS PROBLEM & PYTHON/SQL PIPELINE (RESUME BULLET 1)

"MedTourEasy coordinates cross-border medical travel across 12 source countries, 8 high-acuity specialties like Oncology and Cardiac Surgery, and 30 partner hospitals. Out of 1,500 patient inquiries, 554 completed treatment (36.93%), generating INR 248.3M in revenue and INR 111.5M in gross margin (44.91% margin rate). However, 946 patients failed to convert—including 350 explicitly coded drop-offs representing INR 166.5M in lost case-value exposure.
To diagnose where and why patients leaked, I built a modular Python and SQLite pipeline across 5 tables: patient_journey (1,500 rows), provider_master (30 hospitals), communication_logs (6,025 interactions), treatment_outcomes (554 completed cases), and country_reference (12 countries). Because the SQLite tables stored numbers as TEXT without primary or foreign key enforcement, I wrote automated schema and reconciliation tests. That audit caught a critical funnel flaw: 229 of the 554 completed patients (41.3%) had quote_shared = 0, making naive quote-to-treatment conversion look like 98.75% (554/561). Filtering to strict sequential progression (325/561) revealed the true 57.93% quote conversion rate."

PART 2 (0:45 – 1:25) | COUNTRY-SPECIFIC SLA BREACH & STATISTICAL PROOF (RESUME BULLET 2)

"Second, rather than using a single global response threshold, I joined every inquiry to its country-specific SLA expectation in country_reference (6 hours for Nigeria/Nepal/Sri Lanka, 8 hours for India/UAE/Malaysia, and 12 hours for Bangladesh/Kenya/Oman/Thailand). I discovered that 82.80% of inquiries (1,242 of 1,500) breached SLA, averaging ~17 hours.
Using a Chi-square test of independence, I proved that inquiries meeting SLA booked consultations at 79.46% (205/258) versus 62.72% (779/1,242) when SLA was breached—a 16.74 percentage point drop (Chi-square = 25.78, p = 3.83e-7, Cramer's V = 0.131). I also verified this continuously with Welch's two-sample t-test: booked consultations averaged 15.77 hours response time versus 19.40 hours for non-booked leads (-3.63 hours, t = -7.96, p = 4.84e-15)."

PART 3 (1:25 – 2:10) | LEAKAGE-AWARE ML CLASSIFICATION — 0.8695 ROC-AUC (RESUME BULLET 3)

"Third, to predict treatment_completed early enough for coordinators to intervene, I established a strict 72-hour post-inquiry prediction cutoff. I excluded all post-treatment financials, satisfaction_score (which is structurally null for all 946 non-completed patients), and downstream stage flags (consultation_booked, quote_shared, drop_off_flag), and aggregated communication_logs strictly within <= 72 hours of inquiry.
Using an 80/20 stratified split (1,200 train / 300 test) inside a scikit-learn ColumnTransformer pipeline, I benchmarked balanced Logistic Regression (0.8193 ROC-AUC) against a regularized Random Forest classifier (0.8695 ROC-AUC, 0.7723 Precision, 0.7358 F1). Top drivers were engagement_score (29.6% importance), estimated_case_value_inr (13.0%), intent_score (7.5%), and response_time_hours (7.4%). Because a missed patient costs INR 201k in margin while a follow-up call costs pennies, I tuned the probability threshold from 0.50 to 0.40, boosting Recall from 70.3% to 78.4%."

PART 4 (2:10 – 2:50) | K-MEANS PERSONAS & INR 7.5M MARGIN RECOVERY PLAYBOOK (RESUME BULLET 4)

"Finally, a risk probability alone doesn't separate a low-intent inquiry from an expensive surgical candidate. I standardized intent_score, engagement_score, response_time_hours, and estimated_case_value_inr with StandardScaler and applied K-Means (k=3) to segment patients into: (1) High-Intent Converters (38.1% of cohort, 60.2% completion), (2) High-Risk Disengaged (33.5%, 14.7% completion, slowest 19.8h response), and (3) High-Value At-Risk (28.4%, 426 patients)—who carry the highest average case value at INR 716.8k (2.2x other clusters) but convert at only 31.9%.
Combining our ML risk scores with K-Means personas and our composite Provider Scorecard, we built an operational routing playbook: escalating High-Value At-Risk patients in the 0.35–0.65 probability band to top-performing hospitals (PVD-020, PVD-008) and senior coordinators (COORD-06). Recovering just 10% of the 350 explicit drop-offs (~35 cases) yields INR 16.65M in incremental revenue and INR 7.48M in gross margin."

STRATEGY 3: 10–15 MIN DEEP-DIVE (PART A: BULLETS 1 & 2)

Minutes 0:00 to 8:00 — Architecture, Data Quality, Dual Funnel, Provider Scorecard & SLA Testing

MINUTES 0:00 – 2:00 | BUSINESS CONTEXT, UNIT ECONOMICS & REPOSITORY ARCHITECTURE

- Business Scope: Cross-border medical travel across 12 source countries (India, Bangladesh, UAE, Nigeria, Kenya, Thailand, etc.), 7 acquisition channels, 8 specialties (Cardiac, Oncology, Organ Transplant, Ortho, Fertility, etc.), 30 hospitals, and 10 coordinators.
- Verified Unit Economics: 1,500 Inquiries -> 554 Completed Treatments (36.93%) | Revenue: INR 248,325,245 (INR 448.2k/case) | Service Cost: INR 136,810,889 (INR 247.0k/case) | Realized Gross Margin: INR 111,514,356 (44.91% margin; INR 201.3k/case).
- Non-Completed Breakdown (946 patients / 63.07%): 350 explicitly coded drop-offs (drop_off_flag=1, INR 166.51M case-value exposure) + 596 stagnant open inquiries (212 stuck at Inquiry, 264 at Consultation, 120 at Quote Shared).
- Production Repository: Modular Python (src/data_quality.py, analytics.py, modeling.py, segmentation.py, eda.py) + 11 SQL queries (sql/medtoureasy_analysis.sql) + 6-page Streamlit app (dashboard/app.py) + 11 automated pytest tests.

MINUTES 2:00 – 5:00 | BULLET 1 DEFENSE: DATA QUALITY, DUAL FUNNEL & PROVIDER SCORECARD

1. Schema & Relational Reconciliation Across 5 Datasets (CSV + SQLite mte_patient_intelligence.db):
- Validated 0 PK duplicates/nulls and 0 FK orphans across patient_journey (1,500), provider_master (30), communication_logs (6,025), treatment_outcomes (554), and country_reference (12). Hardened SQLite TEXT columns via explicit CAST(... AS INTEGER/REAL).
- Structural Missingness Classification: satisfaction_score has 946/1,500 nulls (exact 946 non-completed rows); drop_off_stage has 1,150/1,500 nulls (exact non-dropped rows); response_delay_hours has 1,413/6,025 nulls (exact unresponded logs, 23.45%).

2. The '229 Unquoted Completions' Discovery (Why Naive Funnel Math Fails):
- Raw counts: 1,500 Inquiries -> 984 Consultations (65.60%) -> 561 Quotes Shared (57.01% of consults) -> 554 Completions (36.93%).
- Naive division (554/561) implies a 98.75% quote conversion rate! Auditing stage flags proved 229 of 554 completed patients (41.34%) had consultation_booked=1 and treatment_completed=1, but quote_shared=0 (direct package/bypass bookings). Separating Strict Sequential Completions (325/561) reveals the true 57.93% quote-to-treatment conversion rate (21.67% strict overall).

3. Composite Provider Intelligence Scorecard (0–100 Weighted Index across 30 Hospitals, ~50 leads/provider):
- Formula: 30% Norm Conversion + 20% Norm SLA Adherence + 15% Norm Follow-up + 15% Norm Satisfaction + 20% Norm Margin/Lead.
- Top Performers: PVD-020 (Score 80.78 | 52.73% conv | 5.85h resp | 72.73% SLA | INR 85.2k margin/lead) & PVD-008 (Score 71.77 | 48.21% conv | highest margin/lead at INR 111.7k). Bottleneck: PVD-007 (52.0% conv, 18.0% SLA) vs Low: PVD-023 (26.9% conv, 0% SLA).

MINUTES 5:00 – 8:00 | BULLET 2 DEFENSE: SLA BREACH ANALYSIS & STATISTICAL TESTING

1. Country-Specific SLA Benchmark (Why a Global Cutoff Is Flawed):
- Joined patient_journey.response_time_hours against country_reference.baseline_response_expectation_hours: 6h SLA (Nigeria, Nepal, Sri Lanka), 8h SLA (India, Malaysia, Philippines, Tanzania, UAE), and 12h SLA (Bangladesh, Kenya, Oman, Thailand).
- System-Wide SLA Failure: 1,242 / 1,500 inquiries (82.80%) breached SLA; only 258 (17.20%) met SLA. Worst breaches occur in 6h-SLA markets (Nigeria 97.6%, Sri Lanka 97.2%, Nepal 94.7%) because coordinator shifts run on an unsegmented ~17h cycle.

2. Dual Hypothesis Testing Framework (Categorical Chi-Square + Continuous Welch's t-Test):
- Pearson's Chi-Square Test (2x2 Table): SLA Met booked at 79.46% (205/258) vs. SLA Breached at 62.72% (779/1,242) -> -16.74 pp gap (95% CI: [-22.39 pp, -11.09 pp]), Chi2 = 25.7784 (dof=1), p = 3.83e-7 ($p < 0.001$), Cramer's V = 0.1311, Odds Ratio = 0.435.
- Downstream Ripple: SLA compliance also lifts Quote Sharing by +11.47 pp (46.90% vs 35.43%) and Completion by +11.57 pp (46.51% vs 34.94%).
- Welch's Two-Sample t-Test: Booked consultations (n=984) averaged 15.77h (SD 7.91h) vs Non-booked (n=516) at 19.40h (SD 8.63h) -> Delta = -3.63h, Welch t = -7.9569, p = 4.84e-15, Cohen's d = -0.431. Proposed stratified randomized A/B test (<4h routing) for causal proof.

STRATEGY 3: 10–15 MIN DEEP-DIVE (PART B: BULLETS 3 & 4)

Minutes 8:00 to 15:00 — Leakage-Aware ML (0.8695 AUC), Threshold Tuning, K-Means Personas & ROI

MINUTES 8:00 – 12:00 | BULLET 3 DEFENSE: LEAKAGE-AWARE ML PIPELINE (0.8695 ROC-AUC)

1. Prediction Timestamp & Three-Layer Leakage Prevention Architecture (72-Hour Post-Inquiry Cutoff):
- Layer 1 (Post-Treatment Financials & Outcomes): Excluded actual_revenue_inr, service_cost_inr, gross_margin_inr, and all treatment_outcomes fields (length_of_stay_days, complication_flag).
- Layer 2 (Structural Null Proxy — satisfaction_score): satisfaction_score is missing for 100% of non-completed rows (946/946) and 0% of completed rows (0/554). Including it with median imputation lets a tree split on the imputed value for ~1.00 fake AUC. Excluded!
- Layer 3 (Downstream Funnel Flags & Future Comm Logs): Excluded consultation_booked, quote_shared, follow_up_completed, current_stage, drop_off_flag, drop_off_stage, reason_for_drop_off. Filtered communication_logs strictly to 0 <= t_comm - t_inquiry <= 72 hours (engineering early_interactions, early_response_rate, early_avg_response_delay, early_followup_completed).

2. Pipeline Engineering & Model Comparison (Stratified 80/20 Split: 1,200 Train / 300 Test, random_state=42):
- Built scikit-learn ColumnTransformer + Pipeline (SimpleImputer + StandardScaler on 19 numeric features; SimpleImputer + OneHotEncoder on 6 categorical features) fit strictly on X_train.
- Logistic Regression (C=1.0, class_weight='balanced'): ROC-AUC = 0.8193 | Precision = 0.6187 | Recall = 0.7748 | F1 = 0.6880 (TN=136, FP=53, FN=25, TP=86). Calibrated Odds Ratios: engagement_score OR=3.15, intent_score OR=1.63, response_time OR=0.70, case_value OR=0.63.
- Random Forest (n_estimators=500, min_samples_leaf=3, max_features=0.7, class_weight='balanced'): ROC-AUC = 0.8695 (+5.02 pp lift) | Accuracy = 81.33% | Precision = 0.7723 | Recall = 0.7027 | F1 = 0.7358 (TN=166, FP=23, FN=33, TP=78).
- Top 5 RF Features (Gini): engagement_score (29.63%), estimated_case_value_inr (12.96%), intent_score (7.53%), response_time_hours (7.42%), early_avg_response_delay (4.70%).

3. Cost-Sensitive Threshold Optimization (Why 0.50 Is Suboptimal for Medical Tourism):
- False Negative loses ~INR 201,289 in gross margin; False Positive costs ~15 mins of coordinator time. Lowering threshold from 0.50 to 0.40 boosts Recall from 70.27% to 78.38% (+8.11 pp, catching 9 extra completers in test set) with Precision at 67.97% and F1 at 0.7280.

MINUTES 12:00 – 15:00 | BULLET 4 DEFENSE: K-MEANS PERSONAS & FINANCIAL SCENARIOS

1. Unsupervised K-Means Clustering Methodology & Validation:
- Features: intent_score (0–100), engagement_score (0–100), response_time_hours (1–46h), estimated_case_value_inr (INR 75k–1.5M).
- Why StandardScaler Is Mandatory: Without z-score standardization, estimated_case_value_inr (~10^5–10^6 INR) dominates >99.99% of Euclidean distance. Evaluated k in {2..6} via Inertia, Silhouette Score, and Davies-Bouldin Index (k=2 Sil=0.264, k=3 Sil=0.233 / DBI=1.448). Picked k=3 because k=2 failed to isolate high-value surgical patients.

2. The 3 Patient Personas & Operational Routing Playbook:
- High-Intent Converters (Cluster 1 — 571 patients / 38.1%): Intent 75.3 | Engagement 69.7 | Resp 15.0h | Case Value INR 327.0k -> 82.0% consult, 60.2% completion, 10.2% drop-off, INR 52.20M gross margin. Action: Fast-track quotes & automated visa/travel checklists.
- High-Value At-Risk (Cluster 2 — 426 patients / 28.4%): Highest Case Value at INR 716.8k (2.19x Cluster 1!) | Intent 63.1 | Engagement 55.5 | Resp 16.4h -> 67.8% consult, only 31.9% completion, 24.6% drop-off, INR 46.58M margin. Action: #1 Priority! Route leads in 0.35–0.65 ML prob band to top hospitals (PVD-020, PVD-008) & senior coordinators (COORD-06, COORD-03) with itemized financial counseling.
- High-Risk Disengaged (Cluster 0 — 503 patients / 33.5%): Intent 45.6 | Engagement 32.1 | Slowest Resp 19.8h | Case Value INR 402.1k -> 45.1% consult, 14.7% completion, 37.2% drop-off, INR 12.74M margin. Action: Automated <1h WhatsApp/Chatbot intake before human triage.

3. Scenario-Based Financial Opportunity (350 Explicit Drop-Offs = INR 166.51M Case-Value Exposure):
- 5% Recovery (~18 cases): INR 8.33M Rev / INR 3.74M Gross Margin | 10% Recovery (Base Target — ~35 cases): INR 16.65M Rev / INR 7.48M Gross Margin | 15% Recovery (~53 cases): INR 24.98M Rev / INR 11.22M Margin | 20% Recovery (~70 cases): INR 33.30M Rev / INR 14.96M Margin.

BULLET-BY-BULLET INTERVIEWER CROSS-EXAMINATION GUIDE

The Exact Tough Follow-Up Questions Interviewers Will Ask When Reading Each Line of Your Resume

CROSS-EXAMINATION ON BULLET 1 (PYTHON/SQL PIPELINE, 5 DATASETS, FUNNEL & PROVIDERS)

Q1: 'What was the hardest data quality bug you caught across the 5 datasets?'
- Answer: Two critical issues: First, 229 out of 554 completed patients (41.34%) had consultation_booked=1 and treatment_completed=1 but quote_shared=0. Naive funnel math (554/561) reported a 98.75% quote conversion rate; isolating strict sequential completions (325/561) revealed the true 57.93% rate. Second, SQLite stored numeric columns as TEXT without PK/FK constraints, requiring explicit CAST statements across all 11 SQL queries.

Q2: 'Why didn't you rank hospitals purely by conversion rate or total revenue?'
- Answer: Single metrics hide operational bottlenecks. For instance, PVD-007 achieved 52.0% conversion and INR 110.6k margin/lead, but had an 18.0% SLA adherence rate (10.17h response). Conversely, PVD-020 delivered 52.7% conversion with 72.7% SLA adherence (5.85h response). Our 0–100 Provider Intelligence Score weights Conversion (30%), SLA Adherence (20%), Margin/Lead (20%), Follow-up (15%), and Satisfaction (15%).

CROSS-EXAMINATION ON BULLET 2 (16.7 PP SLA CONVERSION GAP & CHI-SQUARE P < 0.001)

Q3: 'Why did you use Chi-square instead of just comparing averages, and what were the exact numbers?'
- Answer: Because both sla_breach (0/1) and consultation_booked (0/1) are binary categorical outcomes, Pearson's Chi-square test of independence is the exact statistical test. SLA-compliant leads converted at 79.46% (205/258) vs. 62.72% (779/1,242) for breached leads (-16.74 pp, 95% CI [-22.39 pp, -11.09 pp], Chi2 = 25.78, p = 3.83e-7). To ensure the binary cutoff wasn't hiding continuous behavior, I also ran Welch's t-test on raw response_time_hours (15.77h booked vs 19.40h unbooked, t = -7.96, p = 4.84e-15).

Q4: 'Does your Chi-square test prove that faster response time CAUSES higher conversion?'
- Answer: No—in observational CRM data, response delay can correlate with medical complexity or arrival time. I framed it as a strong observational association and proposed a stratified randomized A/B test (randomizing leads within country x specialty strata to a <4h rapid-response desk vs standard queue) to measure causal lift.

CROSS-EXAMINATION ON BULLET 3 (LOGISTIC REGRESSION & RANDOM FOREST — 0.8695 ROC-AUC)

Q5: 'Walk me through every feature you dropped to prevent data leakage and why.'
- Answer: I set the prediction point at 72 hours post-inquiry. I dropped: (1) post-treatment financials (actual_revenue_inr, service_cost_inr, gross_margin_inr), (2) satisfaction_score—which is missing for all 946 non-completed rows and populated for all 554 completed rows, so imputing it creates a 100% target leak, (3) downstream funnel flags (consultation_booked, quote_shared, drop_off_flag, current_stage), and (4) communication_logs occurring >72 hours after inquiry.

Q6: 'Why did you keep Logistic Regression if Random Forest achieved 0.8695 vs 0.8193 ROC-AUC?'
- Answer: Random Forest is our production ranking engine (+5.02 pp AUC lift because it captures non-linear interactions like engagement x response_time and case_value x complexity), while Logistic Regression gives calibrated directional Odds Ratios for clinical leadership (e.g., +1 SD in engagement increases completion odds by 3.15x, while +1 SD in case value reduces odds by 0.63x due to price sensitivity).

CROSS-EXAMINATION ON BULLET 4 (K-MEANS SEGMENTATION INTO 3 PATIENT PERSONAS)

Q7: 'Why did you standardize features before K-Means, and why pick k=3 when k=2 had a slightly higher Silhouette score?'
- Answer: K-Means minimizes isotropic Euclidean distance; without StandardScaler, estimated_case_value_inr (ranging up to INR 1.5M) overwhelms 0–100 scores by 10,000x. Across k in {2..6}, k=2 (Silhouette 0.264) only split high vs low engagement. Selecting k=3 (Silhouette 0.233, Davies-Bouldin 1.448) isolated the 426 High-Value At-Risk patients (INR 716.8k avg case value, 31.9% conversion)—the single most actionable segment for senior coordinator escalation.

Q8: 'How do your ML model and K-Means personas work together in production?'
- Answer: They form a 2x2 routing matrix: ML predicts conversion likelihood at 72h, while K-Means identifies economic value and behavioral posture. A High-Value At-Risk patient in the 0.35–0.65 ML probability band gets immediate senior coordinator & top-hospital escalation.